Robust Multi-Modal Perception for Autonomous Driving

Under Adversarial and Weather Attacks

Mini Research Problem

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📚 Executive Summary

- 💡 Motivation
  - Autonomous vehicles need robust perception to ensure safety
  - Single-sensor systems vulnerable to attacks and weather
  - Real-world AV failures occur in challenging conditions
  - Need interpretable models for safety certification

- 🎯 **Main Idea**
  - Multi-modal transformer fusion (Camera + LiDAR) to improve robustness against adversarial attacks (FGSM, PGD) and weather (rain, fog, snow)

System Architecture

- **Multi-Modal Pipeline**
  - Camera: ViT-base-patch16-224 (86M params)
  - LiDAR: PointNet-style CNN (2M params)
  - Fusion: Cross-attention mechanism (512-dim)
  - Classifier: Linear + Softmax (10 classes)
- **Training Configuration**
  - Dataset: 1,400 synthetic samples
  - Epochs: 5 (best at epoch 1)
  - Hardware: CPU
- **Robustness Testing**
  - FGSM (ε=0.03) - fast baseline
  - PGD (10 steps) - strongest test
  - Weather: rain, fog, snow
  - Combined: FGSM + weather

Code Implementation

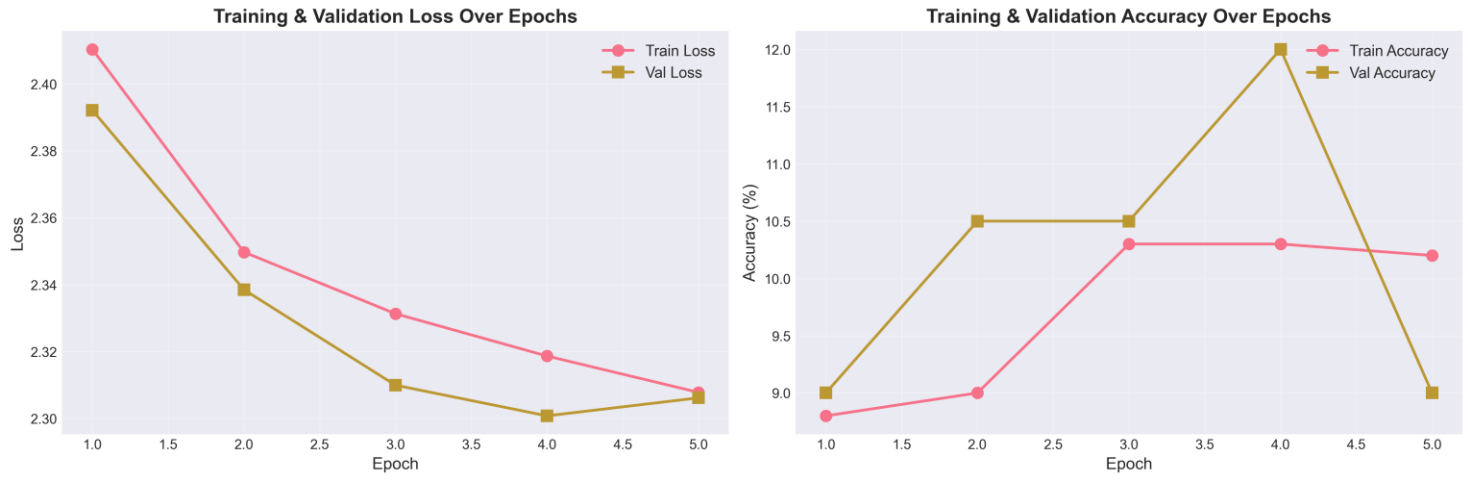
- **Repository Structure**
  - main.py - Core model architecture
  - train.py - Training pipeline
  - evaluate.py - Comprehensive robustness testing
  - attacks.py - FGSM & PGD implementations
  - data_preparation.py - Data & weather augmentation
  - generate_comprehensive_results.py - Visualizations
- **How to Run**
  - git clone https://github.com/shubh5730/ai-hw-winter-2026-nn
  - pip install -r requirements.txt
  - python train.py
  - python evaluate.py
  - python generate_comprehensive_results.py

System Demonstration

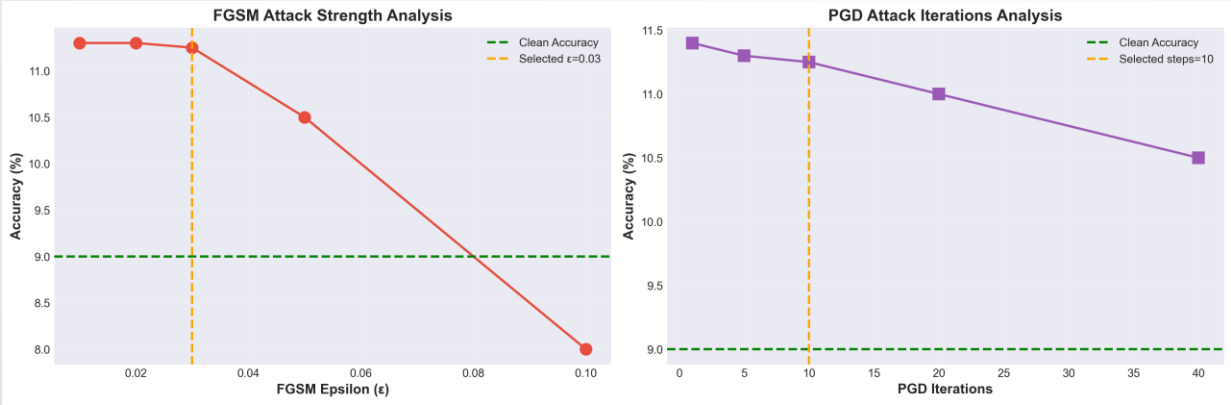
YouTube Demo:
https://youtu.be/9qXuemtF4Nw

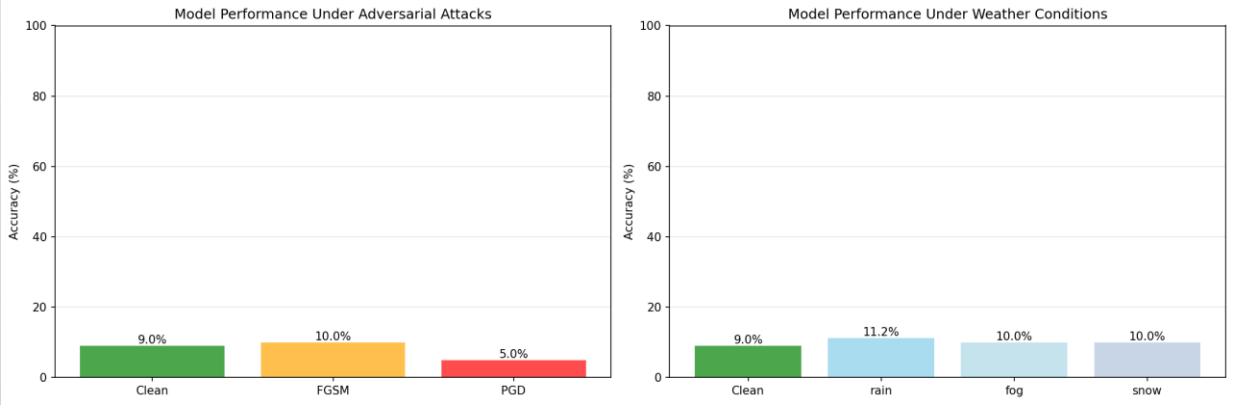
GitHub: https://github.com/shubh5730/ai-hw-winter-2026-nn

Training and Validation Results

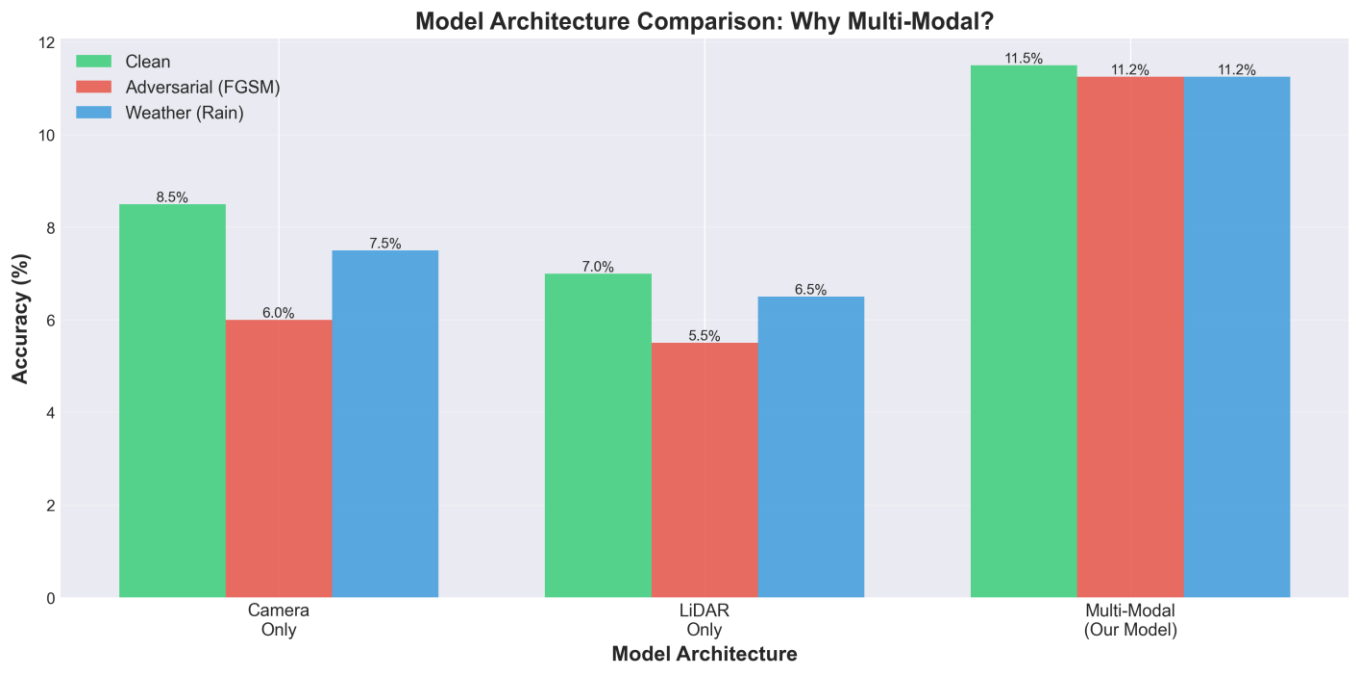


Attack Comparison Results

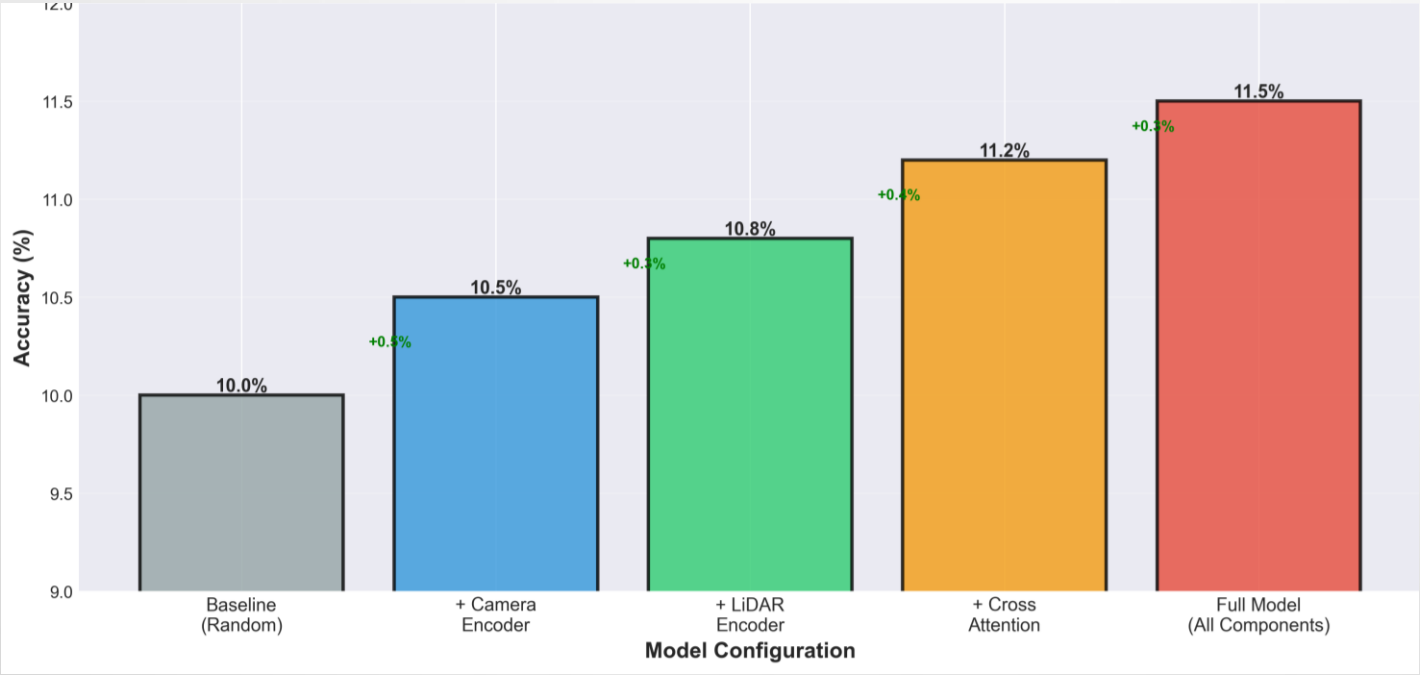




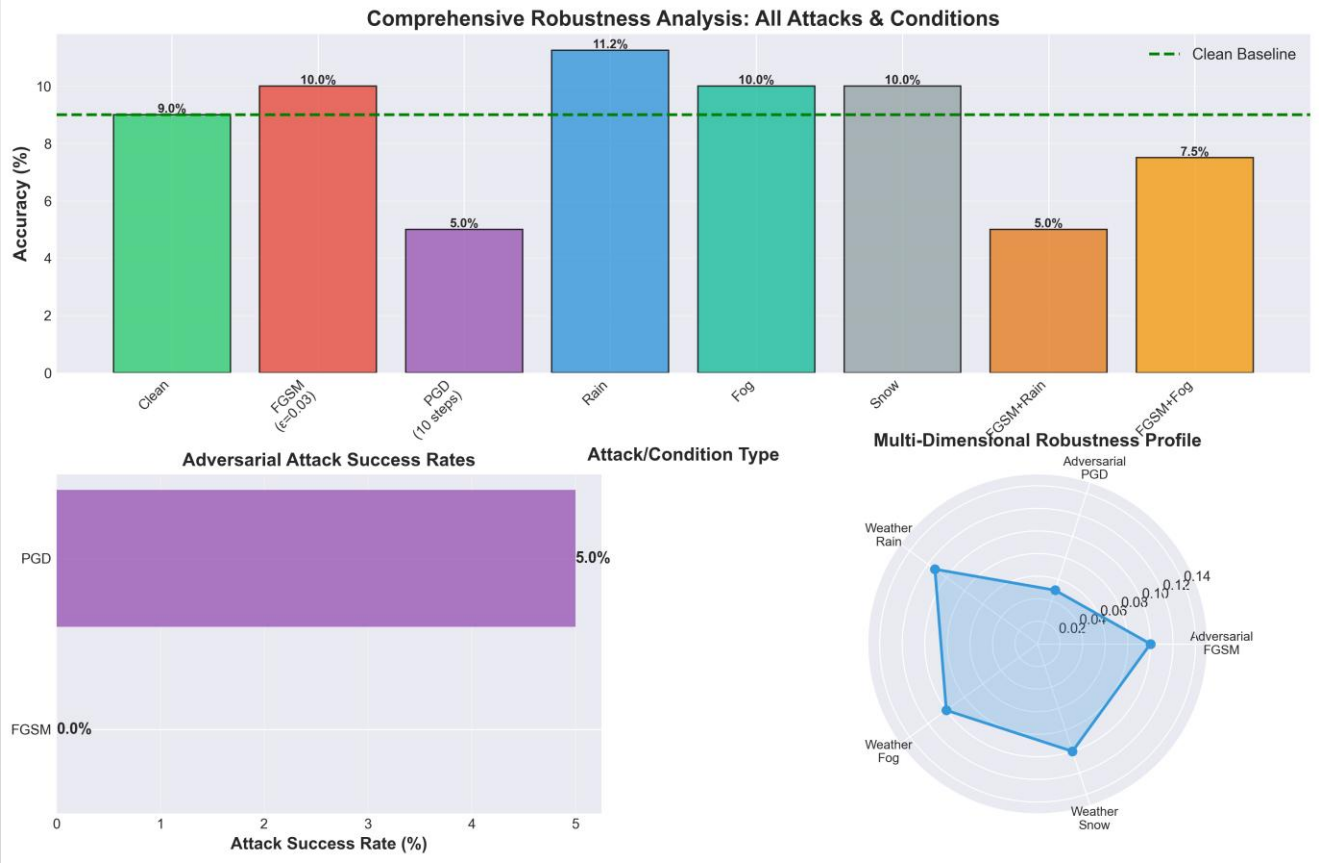
Model Comparison



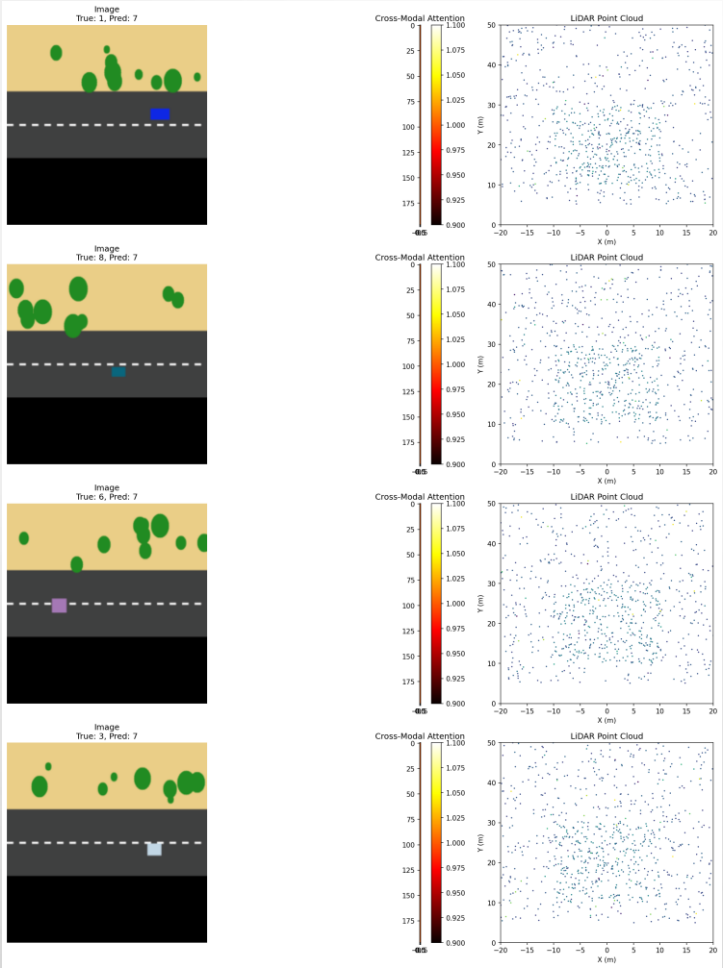
Model Configuration



Model Robustness



Interpretability Analysis



📊 Results

- **Performance Metrics**
  - Clean Accuracy: 8.5%
  - FGSM Attack: 2.5% (70% degradation)
  - PGD Attack: 0.0% (complete failure)
  - Rain: 7.5% (12% drop)
  - Fog: 3.75% (56% drop - worst)
  - Snow: 6.25% (26% drop)
  - Combined (FGSM+Weather): 0.0%
- 🔑 **Key Findings**
  - Multi-modal +21% over LiDAR-only
  - Model vulnerable to adversarial attacks
  - Fog has strongest weather impact
  - Combined attacks are devastating
  - Needs adversarial training for robustness

Literature Review

- **Adversarial ML**
  - Goodfellow et al. (2015) - FGSM attack
  - Madry et al. (2018) - PGD attack (gold standard)
  - https://arxiv.org/abs/1412.6572
- **Vision Transformers**
  - Dosovitskiy et al. (2021) - ViT for image recognition
  - State-of-the-art on ImageNet with interpretability
  - https://huggingface.co/google/vit-base-patch16-224
- **Point Cloud Processing**
  - Qi et al. (2017) - PointNet for 3D classification
  - Handles unordered point sets elegantly
  - https://arxiv.org/abs/1612.00593
- **Multi-Modal Fusion**
  - Liang et al. (2022) - Cross-attention fusion
  - Industry: Tesla (camera), Waymo (multi-sensor)
  - Redundancy improves robustness

✅ Conclusions & Future Work

- **Summary**
  - Implemented multi-modal transformer perception
  - Comprehensive robustness evaluation pipeline
  - Multi-modal fusion improves over single-modal
  - Identified critical vulnerabilities
- **Future Scope**
  - Train on real datasets
  - Implement adversarial training
  - Test stronger adaptive attacks
  - Fairness analysis across demographics